Practice workbook · Show and explain why impulses travel one way across a synapse

Drawing the direction of an impulse across a synapse or through a reflex arc, marking the receptor end, and explaining or naming what keeps transmission one-way.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-SHOW-EXPLAIN-ONE-WAY-IMPULSE

1. Question 6 · part b.iii · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part b.iii only.

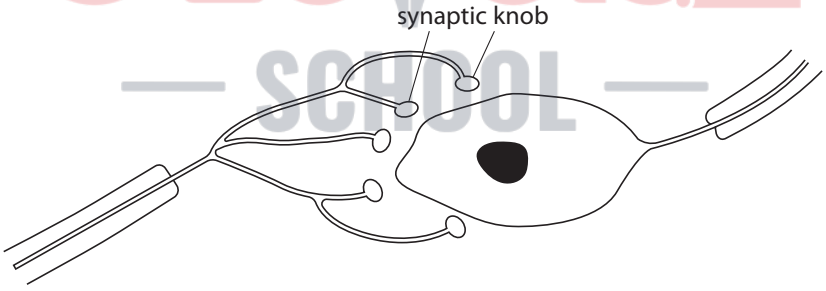
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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1. Question 6 · part b.iii · 4 marks

January 2022 Paper 2

Worked-example retry · Attempt part b.iii only.

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(ii) Explain how an impulse travels across a synapse.

(4)

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(iii) Explain why an impulse can only travel in one direction across a synapse.

(4)

(Total for Question 6 = 12 marks)

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1. Question 6 · part b.iii · 4 marks · Working and answer

Working and answer

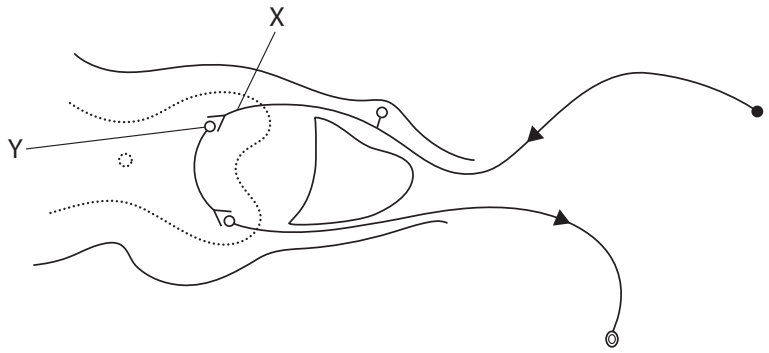
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2. Question 8 · part b · 1 mark

June 2021 Paper 1

Further practice · Attempt part b only.

8 The diagram shows a reflex arc involved in a reflex action.



(a) Describe the function of a reflex action.

(3)

(b) Using a line labelled Z, show on the diagram which part of the reflex arc would be connected to a receptor.

(1)

(c) Describe how the nerve impulse passes from position X to position Y in the reflex arc.

(6)

(Total for Question 8 = 10 marks)

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2. Question 8 · part b · 1 mark · Working and answer

Working and answer

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3. Question 3 · part b.ii · 1 mark

June 2025 Paper 2

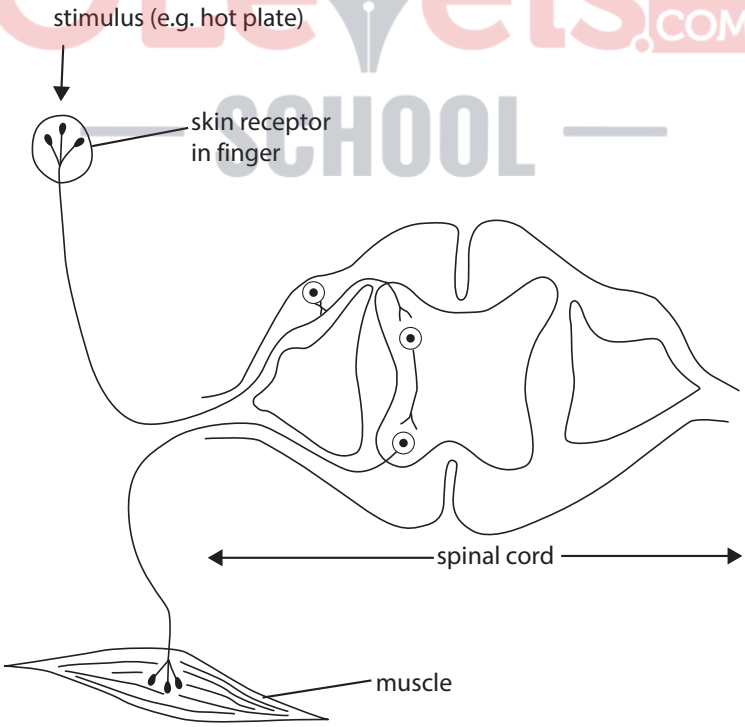
Further practice · Attempt part b.ii only.

3 (a) Describe the meaning of the term **reflex action**.

(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

3. Question 3 · part b.ii · 1 mark

June 2025 Paper 2

Further practice · Attempt part b.ii only.

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(i) Use labels to identify each type of neurone in the diagram.

(3)

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

(1)

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

(1)

.....

(Total for Question 3 = 8 marks)

Working and answer

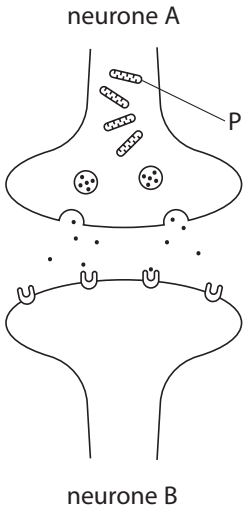
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4. Question 8 · part a.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part a.i only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

(ii) Explain the function of structure P in the transmission of the impulse across the gap. **(3)**

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P 6 2 0 5 4 A 0 2 2 2 4

4. Question 8 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 6 · part b.i · 1 mark

January 2022 Paper 2

Further practice · Attempt part b.i only.

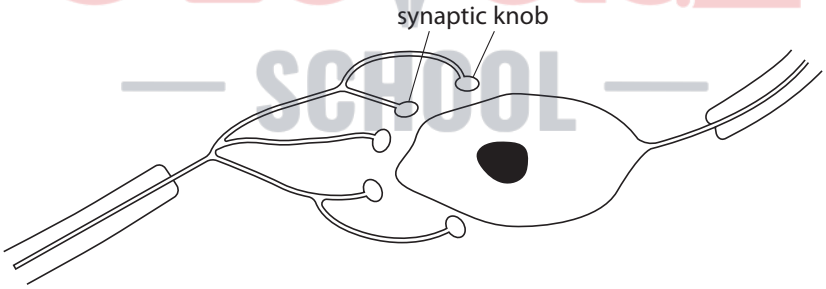
6 (a) Co-ordination of the activities of the body involves the nervous system and the hormonal system.

Complete the table to compare the two systems.

(3)

	Nervous system	Hormonal system
Route of transmission		
Type of signal		
Relative speed		

(b) The diagram shows a synapse.



(i) Draw an arrow on the diagram to show the direction an impulse travels.

(1)

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5. Question 6 · part b.i · 1 mark

January 2022 Paper 2

Further practice · Attempt part b.i only.

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(ii) Explain how an impulse travels across a synapse.

(4)

(iii) Explain why an impulse can only travel in one direction across a synapse.

(4)

(Total for Question 6 = 12 marks)

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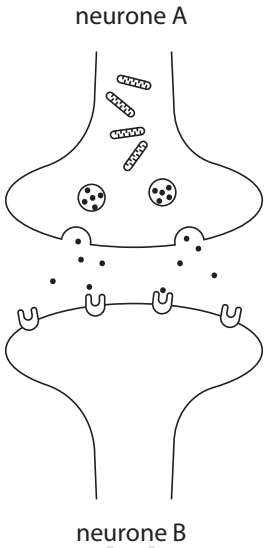
Working and answer

6. Question 3 · part a.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.ii only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

.....

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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P 7 5 9 5 9 A 0 8 2 4

6. Question 3 · part a.ii · 1 mark

November 2024 Paper 2

Further practice · Attempt part a.ii only.

(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

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6. Question 3 · part a.ii · 1 mark · Working and answer

Working and answer

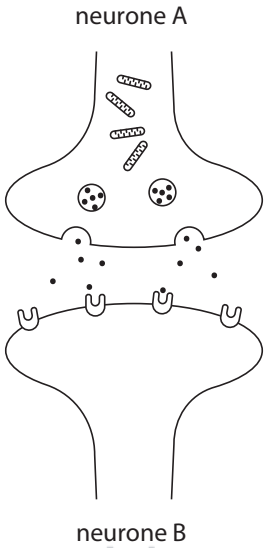
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7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

3 (a) The diagram shows the gap between two neurones, A and B.



(i) Name this gap. **(1)**

(ii) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

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7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

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(iii) Describe how a nerve impulse travels from one side of the gap to the other.

(6)

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(iv) Explain why an impulse can only travel in one direction across the gap.

(3)

7. Question 3 · part a.iv · 3 marks

November 2024 Paper 2

Further practice · Attempt part a.iv only.

(b) Some people take the illegal drug known as cocaine.

State two dangers of taking cocaine.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 13 marks)

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Working and answer

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8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2

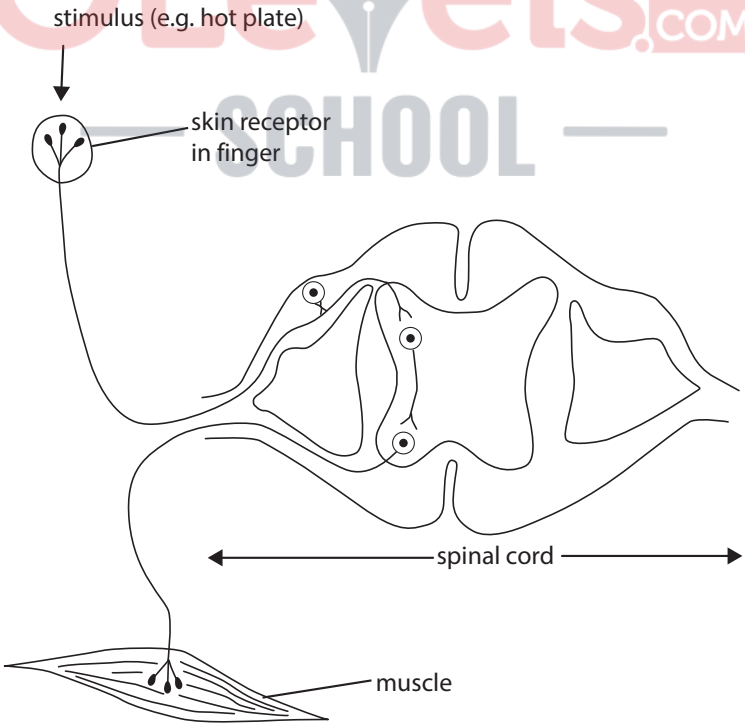
Further practice · Attempt part b.iii only.

3 (a) Describe the meaning of the term **reflex action**.

(3)

(b) The diagram shows a reflex arc.

The reflex arc contains three types of neurone.



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P 7 8 7 6 9 A 0 6 2 0

8. Question 3 · part b.iii · 1 mark

June 2025 Paper 2

Further practice · Attempt part b.iii only.

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(i) Use labels to identify each type of neurone in the diagram.

(3)

(ii) Draw an arrow on the diagram to show the direction of an impulse as it travels through the reflex arc.

(1)

(iii) State which structure in a reflex arc prevents the impulse travelling in the wrong direction.

(1)

.....

(Total for Question 3 = 8 marks)

Working and answer

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